LOGAN THOMAS

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA) | Bachelor of Science, Mechanical Engineering | GPA: 4.0 Expected Jun 2029

SKILLS

Design & Modeling: SolidWorks (CSWA), AutoCAD, Design for Manufacturability (DFM), GD&T / Tolerance Analysis, Rapid Prototyping
Simulation & Analysis: MATLAB, Simulink / Simscape (Model Calibration), Python (Data Analysis)
Data & Machine Learning: scikit-learn (Random Forests), Model Validation (Cross-Validation, ROC-AUC), MATLAB Image Analysis
Manufacturing: Manual Machining, Waterjet, Laser Cutting, Sheet Metal Forming, Silicone Molding, Mechanical Assembly
Electronics & Testing: Circuit Design, Embedded Systems (Arduino), IMU Integration (MPU-6050), PID Control, Design of Experiments (DOE), Root Cause Analysis (RCA), Instrumented Thermal Testing, Soldering

EXPERIENCE

ENGINEER INTERN | DOER Marine | *Manned Research Submersible* Jun 2026 – Present

- Thermal Test Rig & Instrumentation: Designed and built an instrumented test rig to benchmark two battery-wall cooling designs, iterating a foil-and-foam camera enclosure to remove thermal-reflection artifacts across 8 controlled heating trials.
- Thermal Model Calibration: Performed initial calibration of a Simscape/MATLAB heat-transfer model against trial data, tuning the convection coefficient (50 → 70 W/m^{2}K) and adding a neoprene radiation layer to align simulated and measured cooling rates.
- Design for Manufacturability: Created SolidWorks stock-layout assemblies and company-standard procurement drawings for HDPE parts, using custom reference planes to enforce 0.25″ waterjet clearance and minimize material waste.
- Electrical Audit & Documentation: Audited nominal voltage, current, and power specs for all major electrical components against the vehicle schematic, catching misinput voltage errors, and authored the formal test procedure for the thermal study.

ENGINEERING RESEARCH INTERN | I^{2}BL (UCLA Bioelectronics Lab) | *Ferrobotic Viral Testing Platform* Jan 2026 – Present

- Injection Port Integration: Developed a mass-producible molding process embedding custom 2-mm self-sealing RTV silicone ports into testing cartridges, using root-cause analysis of mold geometry and fill sequencing to prevent recurring air-entrapment defects.
- Precision Enclosure: Engineered a modular enclosure with a precision-guided sliding cartridge, achieving sub-millimeter vertical tolerances to preserve optical line-of-sight and unrestricted ferrobot kinematics.
- Optical Subsystem: Re-architected the imaging assembly for interchangeable camera modules, designing a custom diffusion solution to eliminate hotspots and ensure uniform illumination across the microfluidic observation region.

Independent Research Project — Microfluidic Dispenser Characterization & ML Modeling

- End-to-End ML Pipeline: Built a full characterization pipeline end-to-end—experimental design, chip fabrication, 369-trial data collection, and MATLAB image analysis—mapping microfluidic dispenser geometry to droplet output.
- Predictive Modeling & Validation: Trained a Python random-forest model predicting droplet volume to 0.369 µL cross-validated error, diagnosing overfitting via the CV/train ratio and building an inverse-design tool for geometry selection.

TECHNICAL PROJECTS

INTERNAL ARM ENGINEER | X1 Robotics (Student Project Team) | *Autonomous "BB-8" Robot* Sep 2025 – Present

- Yaw Drive Housing: Contributed to an autonomous robot build with a student project team, modeling and assembling a custom yaw motor housing in SolidWorks and optimizing its geometry for 3D-printing manufacturability (DFM).

ELECTRIC GUITAR FABRICATION | Personal Project Jun 2024 – Present

- Precision Manufacturing: Fabricated two custom electric guitars using manual machining (router/bandsaw), achieving precise scale length and fret alignment to guarantee intonation accuracy across the fretboard.
- System Integration: Installed and calibrated a Floyd Rose tremolo bridge, balancing spring tension against string gauge to ensure precise pitch return after mechanical modulation.